

**END TERM EXAMINATION [JULY-2023]
FOURTH SEMESTER [B.TECH]
CIRCUIT AND SYSTEM [EEEC-208]**

Time: 3 Hrs.

Max. Marks: 75

Note: Attempt five questions in all including Q. No. 1 which is compulsory. Select one question from each unit.

Q.1. Attempt all question:

Q.1. (a) What is resonance? Explain series and parallel resonance. (5)

Ans. In an AC electric circuit, when the capacitive reactance is balanced by the inductive reactance at some given frequency, then this condition in the circuit is referred as resonance. The frequency of the supply voltage at which resonance occurs in the circuit is called resonant frequency. At the resonance in the circuit the reactance of the capacitor and inductor cancel each other. Also, at the condition of resonance, no reactive power is taken from the source.

Basis of Difference	Series Resonance	Parallel Resonance
Definition	When a resistor, inductor and capacitor are connected in series across an AC supply and the inductor and capacitor cancel the effect of each other at a particular frequency, then this condition of the series circuit is known as series resonance.	A combination of the resistor, inductor and a capacitor is connected across an AC source and the inductor and capacitor cancel the effect of each other at a specific supply frequency, then this condition of the parallel RLC circuit is known as parallel resonance.
Impedance	The impedance of a series RLC circuit becomes minimum at series resonance.	The impedance of a parallel RLC circuit becomes maximum at parallel resonance.
Admittance	The series RLC circuit offers maximum admittance at series resonance.	The admittance of the parallel RLC circuit at parallel resonance is minimum.
Current	The series resonance result in the maximum current through the circuit.	The current in circuit at parallel resonance is minimum.
Behave of the circuit	The series RLC circuit behaves as an acceptor circuit at series resonance.	The parallel RLC circuit acts as a rejector circuit at parallel resonance.
Magnify	The series resonance magnifies the voltage in the circuit.	The parallel resonance magnifies the current in the circuit.
Equation of effective impedance	The effective impedance of series RLC circuit at series resonance is given by, $Z = R$	The effective impedance of parallel RLC circuit at parallel resonance is given by, $Z = \frac{L}{CR}$

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Quality factor (Q-factor)	For series resonance, the quality factor is given by, $Q - \text{Factor} = \frac{\omega_0 L}{R} = \frac{1}{\omega_0 RC}$	For parallel resonance, the quality factor is given by, $Q - \text{Factor} = \frac{R}{\omega_0 L} = \omega_0 RC$
Applications	The series resonance is widely used in tuning, oscillator circuits, voltage amplifiers, high frequency filters, etc.	The parallel resonance is used in current amplifiers, induction heating, filters, radio-frequency amplifiers, etc.

Q.1. (b) Enlist properties of LTI systems. (5)

Ans. Refer Q. No. 1(b) of First Term Examination 2018 (Pg. No. 1 - 2018).

Q.1. (c) Explain different types of interconnections of two 2-port networks. (5)

Ans. The interconnection between different parameters are shown in Tables below:-

Interconnection	Individual Parameter Matrix	Overall Parameter Matrix
Series-Series or Series	$[Z_a], [Z_b]$	$[Z] = [Z_a] + [Z_b]$
Parallel-Parallel or Parallel	$[Y_a], [Y_b]$	$[Y] = [Y_a] + [Y_b]$
Cascade or Tandem	$[T_a], [T_b]$	$[T] = [T_a] + [T_b]$
Series-Parallel	$[h_a], [h_b]$	$[h] = [h_a] + [h_b]$
Parallel-Series	$[g_a], [g_b]$	$[g] = [g_a] + [g_b]$

UNIT - I

Q.2. (a) Discuss different type of systems. (5)

Ans. System are classified in to the following categories:

- Linear and Non-linear System
- Time variant and Time invariant System
- Linear Time variant and Linear Time invariant System
- Static and Dynamic System
- Casual and Non-casual System
- Invertible and Non-invertible System
- Stable and unstable System

(a) Linear and Non- Linear system:

A system is said to be linear when it satisfies super position and homogenate principles. Consider two system with input as $x_1(t)$, $x_2(t)$, and outputs as $Y_1(t)$, $Y_2(t)$ respectively. Then according to the super position and homogenate principles,

$$T[a_1x_1(t) + a_2x_2(t)] = a_1T[x_1(t)] + a_2T[x_2(t)]$$

$$T[a_1x_1(t) + a_2x_2(t)] = a_1Y_1(t) + a_2Y_2(t)$$

(b) Time Variant and Time Invariant system:

A system is said to be Time variant if its input and output characteristics vary with time. Otherwise, the system is considered as time invariant.

The condition for time invariant system is :

$$y(n, t) = y(n - t)$$

The condition for time variant system is :

$$y(n, t) \neq y(n - t)$$

Where $y(n, t) = T[x(n - t)]$ = input change

$y(n - t)$ = output change

(c) **Linear Time variant(LTV) and Linear Time invariant (LTI) systems:**

If a system is both linear and time variant , then it is called linear time variant (LTV) system.

If a system is both linear and time invariant then that system is called linear time invariant (LTI) system.

(d) **Static and Dynamic systems:**

Static system is memory less where as dynamic system is a memory system.

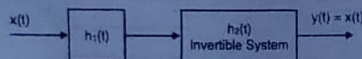
(e) **Casual and non-casual systems:**

A system is said to be casual if its output depends upon present and past inputs, and does not depend upon future inputs.

For non-casual system, the output depends upon future inputs also.

(f) **Invertible and Non-invertible systems:**

A system is said to invertible if the inputs of the system appears at outputs.



$$Y(s) = X(s)H_1(s)H_2(s)$$

$$Y(s) = X(s)H_1(s) \frac{1}{H_1(s)}$$

$$\left(\because H_2(s) = \frac{1}{H_1(s)} \right)$$

$$y(t) = x(t)$$

Hence, the system is invertible.

If $y(t) \neq x(t)$, then the system is said to be non-invertible.

(g) **Stable and Unstable systems:**

The system is said to be stable only when the output is bounded for bounded inputs. For a bounded inputs, if the output is unbounded in the system then it is said to be unstable.

Q.2. (b) Explain Time and Frequency Scaling properties of Z-transform. (6)

Ans. Time Shifting: If we have time shifted sequence such as $x(n - k)$, then its z-transform is given by $Z\{x(n - k)\} = z^{-k} X(z)$

Let's take $n - k = m$, i.e. $n = k + m$ and $y(n) = x(n - k)$. Now here we assuming that $x(n)$ starts from $n = 0$, hence $x(n - k)$ starts from $n = k$, or $n - k = 0$ or from $m = 0$.

$$Y(z) = \sum_{m=0}^{\infty} y[n] z^{-n}$$

$$= \sum_{m=0}^{\infty} x[n - k] z^{-n}$$

$$= \sum_{m=0}^{\infty} x(m) z^{-m-k}$$

$$= z^{-k} \sum_{m=0}^{\infty} x(m) z^{-m}$$

$$= z^{-k} X(z)$$

Scaling in z- domain

When we multiply the signal sequence $x(n)$ in the time domain with an exponential factor a^n , the equivalent z-transform of the new signal is scaled by a factor of a .

Basically, $Z\{a^n x(n)\} = X(z/a)$

Proof of elementary and is shown below,

$$\sum_{n=0}^{\infty} a^n x[n] z^{-n} = \sum_{n=0}^{\infty} x[n] (a^{-1}z)^{-n}$$

$$= \sum_{n=0}^{\infty} x(n) (z/a)^{-n}$$

$$= X(z/a)$$

Q.2. (c) Find the Fourier Transform of the function given below: (5)

$$x(t) = \sin(\omega_0 t) \cos(\omega_0 t)$$

Ans. Given that

$$x(t) = \sin(\omega_0 t) \cos(\omega_0 t)$$

$$= \frac{1}{2} \times 2 \sin(\omega_0 t) \cos(\omega_0 t)$$

$$= \frac{1}{2} \sin 2\omega_0 t$$

$$\sin(2\omega_0 t) = \frac{e^{j2\omega_0 t} - e^{-j2\omega_0 t}}{2j}$$

$$F[\sin(2\omega_0 t)] = F\left[\frac{e^{j2\omega_0 t} - e^{-j2\omega_0 t}}{2j}\right]$$

$$= \frac{1}{2j} (F[e^{j2\omega_0 t}] - F[e^{-j2\omega_0 t}])$$

$$= \frac{1}{2j} [2\pi\delta(\omega - 2\omega_0) - 2\pi\delta(\omega + 2\omega_0)]$$

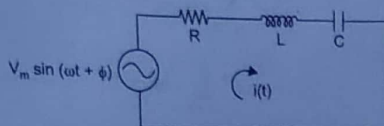
$$= \frac{2\pi}{2j} [\delta(\omega - 2\omega_0) - \delta(\omega + 2\omega_0)] = \frac{\pi}{j} [\delta(\omega - 2\omega_0) - \delta(\omega + 2\omega_0)]$$



OR

Q.3. (a) Write the matrix state equations for series RLC circuit with AC excitation source.

Ans. The series RLC circuit with AC excitation source is shown below



Apply KVL in the circuit, we get

$$V_m \sin(\omega t + \phi) = iR + L \frac{di}{dt} + \frac{1}{C} \int i dt$$

After solving the above differential equation, we get

$$i = e^{at} (C_1 + C_2 t) + \frac{V_m}{\sqrt{R^2 + \left(\frac{1}{\omega C} - \omega L\right)^2}} \sin\left(\omega t + \phi + \tan^{-1} \frac{\left(\frac{1}{\omega C} - \omega L\right)}{R}\right)$$

Q.3. (b) Calculate the Fourier Series coefficients of the continuous-time periodic signal $x(t)$.

$$x(t) = 1 + \sin(\omega_0 t) + 2\cos(\omega_0 t) + \cos(2\omega_0 t + \pi/4)$$

Ans. The given periodic signal is

$$x(t) = 1 + \sin(\omega_0 t) + 2\cos(\omega_0 t) + \cos\left(2\omega_0 t + \frac{\pi}{4}\right)$$

$\sin \omega_0 t$	$2\cos \omega_0 t$	$\cos\left(2\omega_0 t + \frac{\pi}{4}\right)$
$T = \frac{2\pi}{\omega_0}$	$T = \frac{2\pi}{\omega_0}$	$T = \frac{2\pi}{\omega}$

$$x(t) = 1 + \frac{1}{2j} e^{j\omega_0 t} - \frac{1}{2j} e^{-j\omega_0 t} + e^{j\omega_0 t} + e^{-j\omega_0 t} + \frac{1}{2} e^{j\left(2\omega_0 t + \frac{\pi}{4}\right)} + \frac{1}{2} e^{-j\left(2\omega_0 t + \frac{\pi}{4}\right)}$$

$$= a_0 + a_1 e^{j\omega_0 t} + a_{-1} e^{-j\omega_0 t} + a_1 e^{j\omega_0 t} + a_{-1} e^{-j\omega_0 t} + a_2 e^{j\left(2\omega_0 t + \frac{\pi}{4}\right)} + a_{-2} e^{-j\left(2\omega_0 t + \frac{\pi}{4}\right)}$$

$$a_0 = 1; a_1 = 1 - \frac{1}{2}j, a_{-1} = 1 + \frac{1}{2}j;$$

$$a_2 = \frac{\sqrt{2}}{4}(1+j); a_{-2} = \frac{\sqrt{2}}{4}(1-j)$$

Q.3. (c) Explain Time differentiation and Time-integration property of Laplace transform. (5)

Ans. Time differentiation property of Laplace Transform

Statement: The time differentiation property of Laplace transform states that if

$$x(t) \xleftrightarrow{LT} X(s)$$

$$\text{Then, } \frac{d}{dt} x(t) \xleftrightarrow{LT} sX(s) - x(0^-)$$

Proof:

By the definition of Laplace transform, we have,

$$L[x(t)] = \int_0^\infty x(t) e^{-st} dt$$

$$\therefore L\left[\frac{d}{dt} x(t)\right] = \int_0^\infty \left[\frac{dx(t)}{dt}\right] e^{-st} dt$$

$$\therefore \int_a^b u dv = [u \cdot v]_a^b - \int_a^b v \cdot du$$

In this case,

$$u = e^{-st} \text{ and } dv = \left[\frac{dx(t)}{dt}\right] dt$$

$$du = -se^{-st} \text{ and } v = x(t)$$

$$\Rightarrow L\left[\frac{d}{dt} x(t)\right] = \int_0^\infty e^{-st} dx(t) = [e^{-st} x(t)]_0^\infty - \int_0^\infty x(t)(-se^{-st}) dt$$

$$L\left[\frac{d}{dt} x(t)\right] = [0 - x(0^-)] + s \int_0^\infty x(t) e^{-st} dt$$

$$\therefore L\left[\frac{d}{dt} x(t)\right] = sX(s) - x(0^-)$$

or it can be represented as,

$$\frac{d}{dt} x(t) \xleftrightarrow{LT} sX(s) - x(0^-)$$

Integration in Time Domain Property of Laplace Transform:

Statement: The time integration property of Laplace transform states that if

$$x(t) \xleftrightarrow{LT} X(s)$$

Then

$$\int_{-\infty}^t x(\tau) d\tau \xleftrightarrow{LT} \frac{x(s)}{s} + \int_{-\infty}^0 \frac{x(\tau)}{s} d\tau$$

Proof:

Consider a function $y(t)$ as,

$$y(t) = \int_{-\infty}^t x(\tau) d\tau$$

Taking differentiation on both sides with respect to time, we have,

$$\frac{dy(t)}{dt} = x(t) \quad \dots(1)$$

$$\text{Also, } y(0^-) = \int_{-\infty}^0 x(\tau) d\tau \quad \dots(2)$$

Taking the Laplace transform of equation (1), we get,

$$L\left[\frac{dy(t)}{dt}\right] = L[x(t)]$$

$$\Rightarrow s y(s) - y(0^-) = X(s)$$

$$\Rightarrow Y(s) = \frac{X(s)}{s} + \frac{y(0^-)}{s} \quad \dots(3)$$

From eq.(2) and (3), we obtain,

$$Y(s) = \frac{X(s)}{s} + \int_{-\infty}^0 \frac{x(\tau)}{s} d\tau$$

$$\therefore Y(s) = L\left[\int_{-\infty}^t x(\tau) d\tau\right] = \frac{X(s)}{s} + \left[\int_{-\infty}^0 \frac{x(\tau)}{s} d\tau\right]$$

or it can be also represented as,

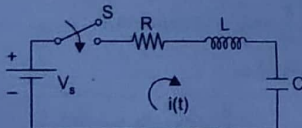
$$\int_{-\infty}^t x(\tau) d\tau \xrightarrow{L} \frac{X(s)}{s} + \int_{-\infty}^0 \frac{x(\tau)}{s} d\tau$$

Thus, it proves the integration in time domain property of the Laplace transform.

UNIT-II

Q.4. (a) Explain transient response of Series RLC circuit with DC excitation.

Ans. Transient response of series RLC circuit with DC excitation is shown below in figure.



Apply KVL, we get

$$L \frac{di(t)}{dt} + Ri(t) + \frac{1}{C} \int i(t) dt = V_s$$

Differentiating both sides, we get

$$L \frac{d^2 i(t)}{dt^2} + R \frac{di(t)}{dt} + \frac{1}{C} i(t) = 0$$

$$\frac{d^2 i(t)}{dt^2} + \frac{R}{L} \frac{di(t)}{dt} + \frac{1}{LC} i(t) = 0 \quad \dots(1)$$

Substituting p^2 for $\frac{d^2 i(t)}{dt^2}$ and p for $\frac{di(t)}{dt}$

$$p^2 + \frac{R}{L} p + \frac{1}{LC} = 0$$

The roots of characteristic equation then

$$p_1, p_2 = \frac{-\frac{R}{L} \pm \sqrt{\left(\frac{R}{L}\right)^2 - 4/LC}}{2}$$

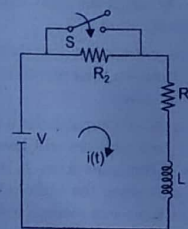
$$\text{Let } \alpha = (-R/2L) \text{ and } \beta = \sqrt{\left(\frac{R}{L}\right)^2 - \frac{1}{LC}}$$

$$\therefore p_1 = \alpha + \beta \text{ and } p_2 = \alpha - \beta$$

Therefore solution of equation (1) becomes

$$i(t) = C_1 e^{p_1 t} + C_2 e^{p_2 t} \quad \dots(8)$$

Q.4. (b) In the network in figure 1, Switch S is closed at $t = 0$, a steady state current having previously been attained. Solve for the current as a function of time using Classical method.



Ans. Steady state current (before the switching action takes place)

$$i(0^-) = \frac{V}{R_1 + R_2}$$

(Since inductor behaves as a short circuit at $t = \infty$)

When switch is closed: R_2 is short circuited.

Applying KVL

$$V = L \frac{di(t)}{dt} + R_1 i(t)$$

$$\text{or } \frac{di(t)}{dt} + \frac{R_1}{L} i(t) = \frac{V}{L}$$

The general solution of the above V equation is given as differential

$$i(t) = \frac{V}{R_1} + K e^{-\frac{R_1}{L} t}$$

$$\text{and } i(0^+) = i(0^-) = \frac{V}{R_1 + R_2}$$

$$\text{or } \frac{V}{R_1 + R_2} = \frac{V}{R_1} + K$$

Table. Relationship for the Parameters

Parameter	Basic Relationship	Voltage-current relationships	Energy
R $G = \frac{1}{R}$	$v(t) = R i(t)$	$v_R(t) = R i_R(t)$ $i_R(t) = G v_R(t)$	$W_R(t) = \int_{-\infty}^t v_R(t) i_R(t) dt$
L	$\psi(t) = L i(t)$	$v_L(t) = L \frac{di_L(t)}{dt}$ $i_L(t) = \frac{1}{L} \int_{-\infty}^t v_L(t) dt$ or $i_L(t) = \frac{1}{L} \int_0^t v_L(t) dt + i_L(0^+)$	$W_L(t) = \frac{1}{2} L i_L^2(t)$
C	$q(t) = C v(t)$	$v_C(t) = \frac{1}{C} \int_{-\infty}^t i_C(t) dt$ or $v_C(t) = \frac{1}{C} \int_0^t i_C(t) dt + v_C(0^+)$ $i_C(t) = C \frac{dv_C(t)}{dt}$	$W_C(t) = \frac{1}{2} C v_C^2(t)$

Therefore, $K = -V \cdot \frac{R_2}{R_1(R_1 + R_2)}$

Hence Solution of our problem is

$$i(t) = \frac{V}{R_1} - \frac{V R_2}{R_1(R_1 + R_2)} e^{-\frac{R_1}{L} t}$$

or

$$i(t) = \frac{V}{R_1} \left(1 - \frac{R_2}{(R_1 + R_2)} e^{-\frac{R_1}{L} t} \right) A$$

OR

Q.5. (a) Define Transient Response. Explain transient and steady state initial conditions and procedure to evaluate these conditions. (7)

Ans. Whenever a circuit is switched from one condition to another, either by a change in the applied source or a change in the circuit elements, there is a transition period during which the branch currents and element voltages change from their former values to new ones. This period is called the *transient*. After the transient has passed, the circuit is said to be in the *steady state*.

Now, the linear differential equation that describes the circuit will have two parts to its solution, the complementary function corresponds to the transient and the particular solution corresponds to the steady-state.

The v-i relation for an inductor or a capacitor is a differential. A circuit containing an inductor L or a capacitor C, and resistors will have current and voltage variables given by differential equations of the same form. It is a linear first order differential equation, with constant coefficients when the values of R, L and C are constant. L and C are storage elements. Circuits have two storage elements like one L and one C are referred to as second order circuits.

Therefore, the series or parallel combinations of R and L or R and C are first order circuits, and RLC in series and RLC in parallel are typical second order circuits.

The circuit changes are assumed to occur at time $t = 0$, and represented by a switch. The switch may be supposed to close/on and open/off at $t = 0$ as shown in figure (a) or (b) respectively, for convenience, it is defined that:

$t = 0^-$, the instant prior to $t = 0$ and

$t = 0^+$, the instant immediately after switching.

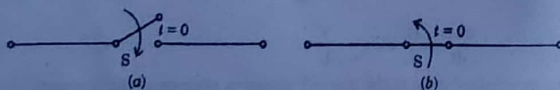


Fig. Switch S is (a) closed at $t = 0$, (b) opened at $t = 0$.

Switching on or off an element or source in a circuit at $t = 0$ will not disturb the storage element so that $i_L(0^-) = i_L(0^+)$ and $v_C(0^-) = v_C(0^+)$. This provides a basis for constructing an equivalent circuit for a charged capacitor voltage (V_0) and current (I_0) carrying inductor.

The voltage-current relationships of the three circuit elements R, L and C are given in Table.

Q.5. (b) In the network shown figure 1, Switch S is closed at $t = 0$, a steady state current having previously been attained. Solve for the current as a function of time using Laplace transform. (8)

Ans. Before the switching action takes place,

$$V = i'(t)(R_1 + R_2) + L \frac{di'(t)}{dt}$$

Taking Laplace transform

$$\frac{V}{s} = I'(s)(R_1 + R_2) + L[sI'(s) - i'(0^+)]$$

Since $i'(0^+) = 0$, therefore

$$I'(s) = \frac{V}{s(R_1 + R_2 + Ls)}$$

$$= \frac{\frac{V}{L}}{s \left(s + \frac{R_1 + R_2}{L} \right)} = \frac{V}{R_1 + R_2} \left[\frac{1}{s} - \frac{1}{s + \frac{R_1 + R_2}{L}} \right]$$

$$i'(t) = \mathcal{F}^{-1}[I'(s)] = \frac{V}{R_1 + R_2} \left[\frac{1}{s} - e^{-\frac{R_1 + R_2}{L} t} \right]$$

Therefore,
$$i'(\infty) = \frac{V}{R_1 + R_2}$$

When switch is closed: R_2 is short circuited. Then

$$V = L \frac{di(t)}{dt} + R_1 i(t)$$

Taking Laplace transform

$$\frac{V}{s} = L[sI(s) - i(0^*)] + R_1 I(s)$$

$$i(0^*) = i'(\infty) = \frac{V}{R_1 + R_2}$$

$$\frac{V}{s} = (sL + R_1)I(s) - \frac{LV}{R_1 + R_2}$$

$$(Ls + R_1)I(s) = \frac{V}{s} + \frac{LV}{R_1 + R_2}$$

$$I(s) = \frac{V}{s(sL + R_1)} + \frac{LV}{(R_1 + R_2)(sL + R_1)}$$

$$\begin{aligned} I(s) &= \frac{V}{R_1} \left[\frac{1}{s} - \frac{1}{s + \frac{R_1}{L}} \right] + \frac{V}{R_1 + R_2} \cdot \frac{1}{s + \frac{R_1}{L}} \\ &= \frac{V}{R_1} \cdot \frac{1}{s} - \frac{1}{s + \frac{R_1}{L}} \left\{ \frac{V}{R_1} - \frac{V}{R_1 + R_2} \right\} \\ &= \frac{V}{R_1} \cdot \frac{1}{s} - \frac{VR_2}{R_1(R_1 + R_2)} \cdot \frac{1}{s + \frac{R_1}{L}} \end{aligned}$$

Therefore

$$i(t) = \mathcal{F}^{-1}[I(s)] = \frac{V}{R_1} - \frac{VR_2}{R_1(R_1 + R_2)} e^{-\frac{R_1}{L} t}$$

or

$$i(t) = \frac{V}{R_1} \left(1 - \frac{R_2}{R_1 + R_2} e^{-\frac{R_1}{L} t} \right)$$

UNIT - III

Q.6. (a) Explain Y-D and D-Y transformation.

Ans.

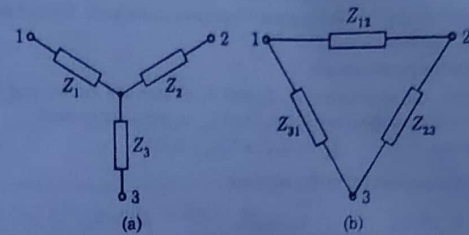


Fig. (a) Star connection, (b) Delta connection.

First, take star connection: The impedance between the terminals 1 and 2 is $Z_1 + Z_2$.

Now, take delta connection: Between the same terminals 1 and 2, there are two parallel paths: one having a impedance of Z_{12} and other having a impedance of $(Z_{23} + Z_{31})$. Therefore,

Impedance between terminals 1 and 2 is

$$Z_{12} \parallel (Z_{23} + Z_{31}) = \frac{Z_{12} \cdot (Z_{23} + Z_{31})}{Z_{12} + (Z_{23} + Z_{31})}$$

As impedance between terminals 1 and 2 have to be the same. Therefore,

$$Z_1 + Z_2 = \frac{Z_{12} \cdot (Z_{23} + Z_{31})}{Z_{12} + Z_{23} + Z_{31}} \quad \dots(1)$$

Similarly, for terminals 2 and 3 and terminals 3 and 1, we get

$$Z_2 + Z_3 = \frac{Z_{23} \cdot (Z_{31} + Z_{12})}{Z_{12} + Z_{23} + Z_{31}} \quad \dots(2)$$

and

$$Z_3 + Z_1 = \frac{Z_{31} \cdot (Z_{12} + Z_{23})}{Z_{12} + Z_{23} + Z_{31}} \quad \dots(3)$$

Delta to Star Conversion

Impedances Z_{12} , Z_{23} and Z_{31} in delta are known and it is required to calculate equivalent impedances Z_1 , Z_2 and Z_3 in star connection.

For this, now subtracting equation (2) from equation (1) and adding the result to equation (3), we get

$$Z_1 = \frac{Z_{12} Z_{31}}{Z_{12} + Z_{23} + Z_{31}} \quad \dots(4)$$

Similarly, we can determine

$$Z_2 = \frac{Z_{23} Z_{12}}{Z_{12} + Z_{23} + Z_{31}} \quad \dots(5)$$

and

$$Z_3 = \frac{Z_{31} Z_{23}}{Z_{12} + Z_{23} + Z_{31}} \quad \dots(6)$$

Therefore, the equivalent impedance of each arm of the star is given by the product of the impedance of the two delta sides that meet at its ends divided by the sum of the three delta impedances.

Star to Delta conversion:

In this case, the impedance Z_1 , Z_2 and Z_3 in star are known and it is required to calculate equivalent impedance Z_{12} , Z_{23} and Z_{31} in delta connection.

Let us define $Z_s = Z_1 Z_2 + Z_2 Z_3 + Z_3 Z_1$

Then from equation (4) to (6), we have

$$Z_s = \frac{Z_{12} Z_{23} Z_{31}}{(Z_{12} + Z_{23} + Z_{31})^2} + \frac{Z_{12} Z_{23} Z_{31}}{(Z_{12} + Z_{23} + Z_{31})^2} + \frac{Z_{12} Z_{23} Z_{31}}{(Z_{12} + Z_{23} + Z_{31})^2}$$

$$\text{or } Z_s = \frac{Z_{12} Z_{23} Z_{31}}{Z_{12} + Z_{23} + Z_{31}} \quad \dots(7)$$

From equation (7) and (4), we have

$$Z_s = Z_{12} Z_3 \text{ or } Z_{12} = \frac{Z_s}{Z_3}$$

$$\text{or } Z_{12} = \frac{Z_1 Z_2 + Z_2 Z_3 + Z_3 Z_1}{Z_3} = Z_1 + Z_2 + \frac{Z_1 Z_2}{Z_3} \quad \dots(8)$$

Similarly

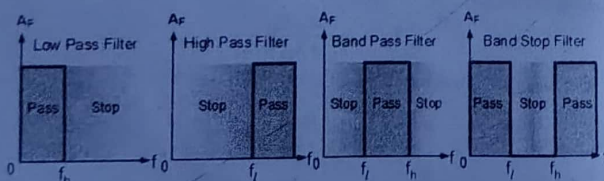
$$Z_{23} = \frac{Z_1 Z_2 + Z_2 Z_3 + Z_3 Z_1}{Z_1} = Z_2 + Z_3 + \frac{Z_2 Z_3}{Z_1} \quad \dots(9)$$

$$\text{and } Z_{31} = \frac{Z_1 Z_2 + Z_2 Z_3 + Z_3 Z_1}{Z_2} = Z_3 + Z_1 + \frac{Z_3 Z_1}{Z_2} \quad \dots(10)$$

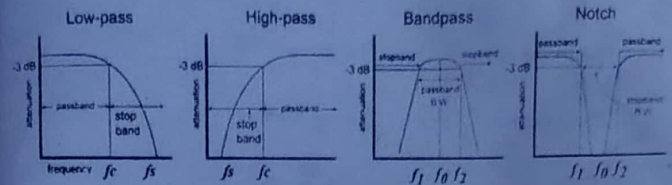
Therefore, the equivalent impedance of each arm of delta is given by the sum of star impedance between those terminals plus the product of these two star impedance divided by the third star impedance.

Q.6. (b) Explain all the passive filter circuits and their frequency response for ideal and practical circuits.

Ans. The ideal response curve are shown below



The practical response curve are shown below

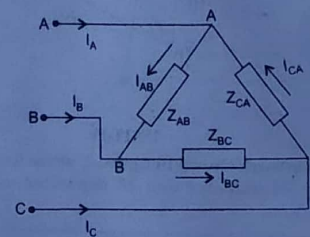


Q.7. (a) Discuss Thevenin's and Norton's Theorems.

Ans. Refer Q.No. 12 (page no. 10-2021)

Q.7. (b) Three impedance $Z_{AB} = (10 + j0) \Omega$, $Z_{BC} = (8 + j6) \Omega$ and $Z_{CA} = (5 - j5) \Omega$ are delta connected to a symmetrical 3-phase 400 V 50 Hz supply of phase sequence ABC. Calculate phase and the line current and total active power consumed. Also draw phasor diagrams. (11)

Ans. The delta connected circuit is given below as



$$I_{AB} = \frac{V}{Z_p} = \frac{400}{\sqrt{(10)^2}} = \frac{400}{10} = 40 \text{ A}$$

$$I_{BC} = \frac{400}{\sqrt{8^2 + 6^2}} = \frac{400}{10} = 40 \text{ A}$$

$$I_{CA} = \frac{400}{\sqrt{5^2 + 5^2}} = \frac{400}{5\sqrt{2}} = \frac{100}{\sqrt{2}} = \frac{80}{\sqrt{2}} \text{ A}$$

Line current are obtained by applying KCL at junction A, B and C. Therefore

$$I_A = I_{AB} - I_{CA} = \left(40 - \frac{80}{\sqrt{2}} \right) \text{ A}$$

$$I_B = I_{BC} - I_{AB} = 40 - 40 = 0 \text{ A}$$

$$I_C = I_{CA} - I_{BC} = \left(\frac{80}{\sqrt{2}} - 40 \right) \text{ A}$$

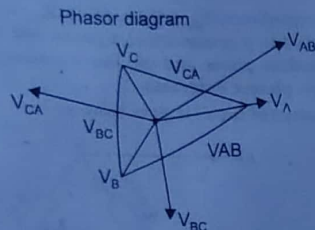
Phase current

$$= \frac{I_L}{\sqrt{3}}$$

$$(i) \text{ Phase current} = \frac{I_A}{\sqrt{3}} = \frac{\left(40 - \frac{80}{\sqrt{2}}\right)}{\sqrt{3}} \text{ A}$$

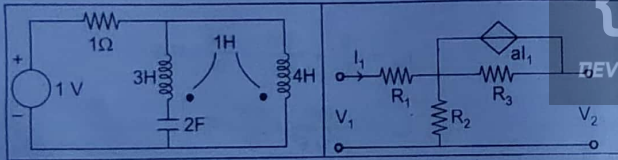
$$(ii) \text{ Phase current} = \frac{I_B}{\sqrt{3}} = \frac{0}{\sqrt{3}} = 0 \text{ A}$$

$$(iii) \text{ Phase current} = \frac{I_C}{\sqrt{3}} = \frac{\left(\frac{80}{\sqrt{2}} - 40\right)}{\sqrt{3}} \text{ A}$$



UNIT-IV

Q.8. (a) For the network shown in figure 2, write f-cut set and f-loop matrix for the tree having 3H inductor and 2F capacitor as twigs. Also write the matrix cut-set equations.



Ans. Since the dots are on the same side, the mutual inductance would be positive. The graph of the network is shown in figure (a), taking branches 2(3H) and 4(2F) as twigs. The f-cut set matrix, Cutsets/Branches

$$Q_f = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & -1 & 1 & 1 & 0 \\ 4 & -1 & 0 & 1 & 1 \end{bmatrix}$$

$$Y_b = Z_b^{-1} = \begin{bmatrix} [1]^{-1} & 0 & 0 & 0 \\ 0 & [3s \ s]^{-1} & 0 \\ 0 & [s \ 4s] & 0 \\ 0 & 0 & 0 & [1/2s]^{-1} \end{bmatrix}$$



$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \frac{4}{11s} & -\frac{1}{11s} & 0 \\ 0 & -\frac{1}{11s} & \frac{3}{11s} & 0 \\ 0 & 0 & 0 & 2s \end{bmatrix}$$

$$Q_f Y_b = \begin{bmatrix} -1 & \frac{3}{11s} & \frac{2}{11s} & 0 \\ -1 & -\frac{3}{11s} & \frac{3}{11s} & 2s \end{bmatrix}$$

$$Y_t = Q_f Y_b Q_f^T = \begin{bmatrix} -1 & \frac{3}{11s} & \frac{2}{11s} & 0 \\ -1 & -\frac{3}{11s} & \frac{3}{11s} & 2s \end{bmatrix} \begin{bmatrix} -1 & -1 \\ 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 + \frac{5}{11s} & 1 + \frac{2}{11s} \\ 1 + \frac{2}{11s} & 1 + \frac{3}{11s} + 2s \end{bmatrix}$$

$$I = Q_f Y_b V_s - Q_f I_s$$

$$= Q_f Y_b V_s \text{ (Since } I_s = 0)$$

$$= \begin{bmatrix} -1 & \frac{3}{11s} & \frac{2}{11s} & 0 \\ -1 & -\frac{3}{11s} & \frac{3}{11s} & 2s \end{bmatrix} \begin{bmatrix} -1 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Hence, the matrix cut-set equation is

$$I = Y_t V_t$$

or

$$\begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 + \frac{5}{11s} & 1 + \frac{2}{11s} \\ 1 + \frac{2}{11s} & 1 + \frac{3}{11s} + 2s \end{bmatrix} \begin{bmatrix} V_2 \\ V_4 \end{bmatrix}$$

Q.8. (b) Find the hybrid parameters for the network shown in figure 3.

Ans. Case I: As we know

$$h_{11} = \left. \frac{V_1}{I_1} \right|_{V_2=0}$$

and

$$h_{21} = \left. \frac{I_2}{I_1} \right|_{V_2=0}$$

Therefore, short circuiting the output port as shown in figure (a) Let v be the node voltage at the junction of R_1 , R_2 and R_3 . Then

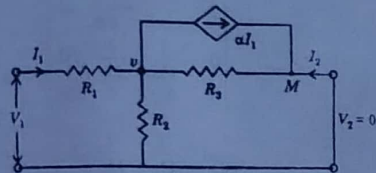


Fig. (a)

Applying KCL at the node;

$$I_1 = \frac{v}{R_2} + \frac{v}{R_3} + \alpha I_1$$

This gives

$$v = \frac{(1-\alpha)I_1 R_2 R_3}{R_2 + R_3}$$

Applying KVL:

$$V_1 = I_1 R_1 + v$$

or

$$V_1 = I_1 \left[R_1 + \frac{(1-\alpha)R_2 R_3}{R_2 + R_3} \right]$$

Now, again applying KCL at node M;

$$\frac{v}{R_3} + \alpha I_1 + I_2 = 0$$

$$I_2 = -\alpha I_1 - \frac{1}{R_3} \left[\frac{(1-\alpha)I_1 R_2 R_3}{R_2 + R_3} \right]$$

$$= I_1 \left[-\alpha - \frac{(1-\alpha)R_2}{R_2 + R_3} \right] = -I_1 \left[\frac{\alpha R_3 + R_2}{R_2 + R_3} \right]$$

So,

$$h_{11} = \left[R_1 + \frac{(1-\alpha)R_2 R_3}{R_2 + R_3} \right] \text{ and } h_{21} = - \left[\frac{\alpha R_3 + R_2}{R_2 + R_3} \right]$$

Case II:

$$h_{12} = \left. \frac{V_1}{V_2} \right|_{I_1=0} \text{ and } h_{22} = \left. \frac{I_2}{V_2} \right|_{I_1=0}$$

Therefore, input circuiting the input port as shown in figure b (current source is open circuited since controlled current source is dependent on input current I_1 which is zero).

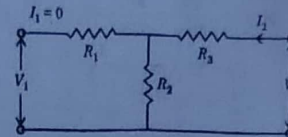


Fig. (b)

$$V_2 = I_2(R_2 + R_3)$$

and

$$V_1 = I_2 R_2$$

So

$$h_{12} = \frac{R_2}{R_2 + R_3}$$

and

$$h_{22} = \frac{1}{R_2 + R_3}$$

OR

Q.9. (a) What is Hurwitz Polynomial? Explain properties of Hurwitz polynomial. (4)

Ans. Hurwitz Polynomial:

Another element of realizability is a class of polynomial known as Hurwitz polynomial which is, in fact, the denominator polynomial of the network function satisfying certain conditions.

In Brief, A polynomial $P(s)$ is said to be Hurwitz if the following conditions are satisfied:

- (1) $P(s)$ is real when s is real.
- (2) the roots of $P(s)$ have real parts which are zero or negative.

Properties of the Hurwitz polynomial:

As a result of the above conditions (1) and (2), Hurwitz polynomial $P(s)$ have the following properties:

- (i) If the polynomial $P(s)$ can be written as

$$P(s) = a_n s^n + a_{n-1} s^{n-1} + \dots + a_1 s + a_0$$

then, All the coefficient a_i must be positive. A corollary is that between the highest order term in s and the lowest order term, none of the coefficients may be zero unless the polynomial is even or odd. In the other words, $a_{n-1}, a_{n-2}, \dots, a_2, a_1$.

- (ii) Both the odd and even parts of a Hurwitz polynomial $P(s)$ have roots on the $j\omega$ -axis only. If we denote the even part of the $P(s)$ as $M(s)$ and the odd part as $N(s)$, so that

$$P(s) = M(s) + N(s)$$

then $M(s)$ and $N(s)$ both have roots on the $j\omega$ -axis

- (iii) As a result of property (ii), if $P(s)$ is either even or odd, all its roots are on the $j\omega$ -axis (including origin).

(iv) The continued fraction expansion of the ratio $\psi(s)$ of the odd to even parts ($N(s)/M(s)$) or the even to odd parts ($M(s)/N(s)$) of a Hurwitz polynomial yields all positive quotients terms. As,

$$\psi(s) = \frac{N(s)}{M(s)} \text{ or } \frac{M(s)}{N(s)} \\ = q_1 s + \frac{1}{q_2 s + \frac{1}{q_3 s + \dots + \frac{1}{q_n s}}}$$

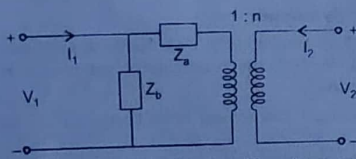
where the quotients $q_1, q_2, \dots, q_n$ must be positive if the polynomial $P(s) = M(s) + N(s)$ is Hurwitz.

(v) If $P(s)$ is Hurwitz polynomial and $W(s)$ is a multiplicative factor. Then $P_1(s) = P(s)W(s)$ is also Hurwitz polynomial, if $W(s)$ is Hurwitz polynomial.

(vi) In case the polynomial is either only even or only odd, it is not possible to obtain the continued fraction expansion in such cases, the polynomial $P(s)$ is Hurwitz if the ratio of $P(s)$ and its derivative $P'(s)$ gives a continued fraction expansion.

Note: Generally, we can check the given polynomial is Hurwitz or not, by using properties (i) and (iv)/(v)/(vi) for expansions (1) and (2) respectively.

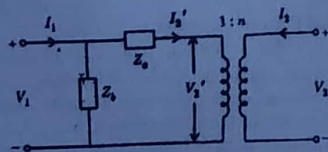
Q.9. (b) For the network shown in figure 4, find Z, Y, h, g, T and T' parameters.



Ans. Let the V_2' and I_2' be the input voltage and input current of the transformer respectively, as shown in figure below. Then the loop equation become

$$V_1 = (I_1 - I_2')Z_b$$

$$V_2 = nV_2' = n(V_1 - I_2'Z_b)$$



and

$$I_2 = -\frac{1}{n}I_2'$$

or

$$I_2' = -nI_2$$

Therefore,

$$V_1 = (I_1 + nI_2)Z_b$$

or

$$V_1 = Z_b I_1 + nZ_b I_2 \quad \dots(A)$$

and

$$V_2 = n(V_1 + nI_2 Z_b)$$

or

$$V_2 = n(I_1 Z_b + nI_2 Z_b + nI_2 Z_b)$$

or

$$V_2 = nZ_b I_1 + n^2(Z_b + Z_b) I_2 \quad \dots(B)$$

from equations (A) and (B);

$$(a) \quad Z_{11} = \left. \frac{V_1}{I_1} \right|_{I_2=0} = Z_b; \quad Z_{21} = \left. \frac{V_2}{I_1} \right|_{I_2=0} = nZ_b$$

$$Z_{12} = \left. \frac{V_1}{I_2} \right|_{I_1=0} = nZ_b; \quad Z_{22} = \left. \frac{V_2}{I_2} \right|_{I_1=0} = n^2(Z_b + Z_b)$$

and,

$$[Y] = [Z]^{-1} = \begin{bmatrix} Z_b & nZ_b \\ nZ_b & n^2(Z_b + Z_b) \end{bmatrix}^{-1}$$

Therefore,

$$Y_{11} = \frac{Z_b + Z_b}{Z_b Z_b}; \quad Y_{12} = -\frac{1}{nZ_b}$$

$$Y_{21} = -\frac{1}{nZ_b}; \quad Y_{22} = \frac{1}{n^2 Z_b}$$

$$A = \left. \frac{V_1}{V_2} \right|_{I_2=0} = \frac{1}{n}; \quad C = \left. \frac{I_1}{V_2} \right|_{I_2=0} = \frac{1}{nZ_b}$$

$$B = \left. \frac{V_1}{-I_2} \right|_{V_2=0} = nZ_b; \quad D = \left. \frac{I_1}{-I_2} \right|_{V_2=0} = n \left(\frac{Z_b + Z_b}{Z_b} \right)$$

$$(\Delta T = AD - BC = 1)$$

And,

$$A' = \frac{D}{\Delta T} = n \frac{(Z_b + Z_b)}{Z_b}$$

$$B' = \frac{B}{\Delta T} = n Z_b$$

$$C' = \frac{C}{\Delta T} = \frac{1}{n Z_b}$$

$$D' = \frac{A}{\Delta T} = \frac{1}{n}$$

(c).

$$h_{11} = \left. \frac{V_1}{I_1} \right|_{V_2=0} \left(\equiv \frac{1}{Y_{11}} \right) = \frac{Z_b Z_b}{Z_b + Z_b}$$

$$h_{21} = \left. \frac{I_2}{I_1} \right|_{V_2=0} = \left(-\frac{1}{D} \right) = -\frac{1}{n} \left(\frac{Z_b}{Z_a + Z_b} \right)$$

$$h_{12} = \left. \frac{V_1}{V_2} \right|_{I_1=0} = \left(\frac{1}{A'} \right) = \frac{Z_b}{n(Z_a + Z_b)}$$

$$h_{22} = \left. \frac{I_2}{V_2} \right|_{I_1=0} = \left(\frac{1}{Z_{22}} \right) = \frac{1}{n^2(Z_a + Z_b)}$$

And,

$$[g] = [h]^{-1}$$

$$\Delta h = h_{11}h_{22} - h_{12}h_{21}$$

$$= \frac{Z_a Z_b}{n^2(Z_a + Z_b)^2} + \frac{Z_b^2}{n^2(Z_a + Z_b)^2} = \frac{Z_b}{n^2(Z_a + Z_b)}$$

Therefore,

$$g_{11} = \frac{h_{22}}{\Delta h} = \frac{1}{Z_b}; \quad g_{21} = -\frac{h_{12}}{\Delta h} = n$$

$$g_{12} = -\frac{h_{21}}{\Delta h} = -n; \quad g_{22} = \frac{h_{11}}{\Delta h} = n^2 Z_a$$

**MID TERM EXAMINATION [APRIL-2024]
FOURTH SEMESTER [B.TECH]
CIRCUITS AND SYSTEMS [EEEC-208]**

Time: 1:30 Hrs.

Max. Marks: 30

Note: Attempt Q. No. 1 which is compulsory and any two more questions from remaining.

Q.1. (a) Check linearity and time-invariance of system $y(n) = 2n + x(n)$. (2)

Ans. $y(n) = 2n + x(n)$

$$x_1(n) \rightarrow y_1(n) = 2n + x_1(n)$$

$$x_2(n) \rightarrow y_2(n) = 2n + x_2(n)$$

$$x_3(n) \rightarrow y_3(n) = 2n + x_3(n)$$

$$x_3(n) = ax_1(n) + bx_2(n)$$

a and b are arbitrary scalars.

$$y_3(n) = ay_1(n) + by_2(n)$$

$$y_3(n) = 2n + x_3(n) = 2n + [ax_1(n) + bx_2(n)]$$

$$= 2n + ax_1(n) + bx_2(n)$$

L.H.S $\neq$ R.H.S Not Linear System.

$$x_2(n) = x_1(n - n_0) \quad x_2(n) \text{ is obtained by shifting in } x_1(n) \text{ in time.}$$

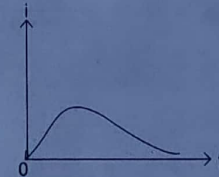
$$y_2(n) = y_1(n - n_0)$$

$$y_2(n) = 2n + x_2(n) = 2n + x_1(n - n_0)$$

$$y_2(n) = y_1(n - n_0) \text{ is a Time Invariant system.}$$

Q.1. (b) Explain an overdamped response of series RLC circuit and sketch the curve. (2)

Ans.



Current over damping for RLC series circuit with DC excitation (2nd order circuit).

Q.1. (c) Find the Laplace transform and associated ROC of signal $x(t) = e^{-2t} [u(t) - u(t-5)]$. (3)

Ans.

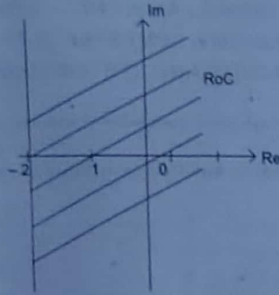
$$x(t) = e^{-2t} [u(t) - u(t-5)]$$

$$x(t) = e^{-2t} u(t) - e^{-2t} u(t-5)$$

Take Laplace Transform on both sides,

$$e^{-2t} u(t) \xrightarrow{LT} \frac{1}{S+2} \quad S > -2$$

$$e^{-2t} u(t-5) \xrightarrow{LT} \frac{e^{-5S}}{(S+2)} \quad S > -2$$



Q.1. (d) For the LTI system described by the following equation. Determine the impulse function of system. (3)

$$y''(t) + y'(t) - 2y(t) = x(t)$$

Ans. $y''(t) + y'(t) - 2y(t) = x(t)$

Take Laplace Transform on both sides

$$S^2 Y(S) + SY(S) - 2Y(S) = X(S)$$

$$Y(S) [S^2 + S - 2] = X(S)$$

$$\frac{Y(S)}{X(S)} = \frac{1}{S^2 + S + 2}$$

$$H(S) = \frac{Y(S)}{X(S)} = \frac{1}{(S-1)(S+2)} = \frac{K_1}{S-1} + \frac{K_2}{S+2}$$

$$K_1 = (S-1) \times \frac{1}{(S-1)(S+2)} \Big|_{S=1} = \frac{1}{3}$$

$$K_2 = \frac{(S+2)}{(S-1)(S+2)} \Big|_{S=-2} = \frac{-1}{3}$$

$$H(S) = \frac{1}{3(S-1)} - \frac{1}{3(S+2)}$$

Take Inverse LT

$$h(t) = [3e^t + 3e^{-2t}] u(t). \quad \text{Ans.}$$

Q.2. (a) Given $X(z) = \frac{z(z-4)}{(z-1)(z-2)(z-3)}$

(i) State all the possible regions of convergence.

(ii) For which ROC, z-transform $X(z)$ is causal sequence. (6)

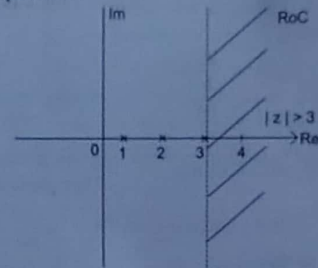
Ans. If we see the all roots of the given convergence is

$$X(z) = \frac{z(z-4)}{(z-1)(z-2)(z-3)}$$

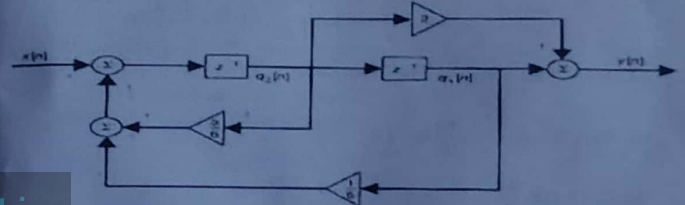
$$z = 0, 1, 2, 3, 4.$$

$$(i) 0 < |z| < 1, \quad 1 < |z| < 2, \quad 2 < |z| < 3, \quad |z| > 3$$

(ii) $X(z)$ is causal sequence for



Q.2. (b) Find the state space representation of the discrete-time system with the state variables $q_1(n)$ and $q_2(n)$ shown in Figure. (4)



Ans. Matrix form

$$q'_1(t) = -q_1(t) + x(t), \quad q'_2(t) = -2q_2(t) - x(t)$$

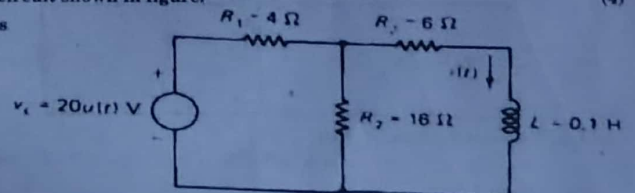
$$q'_3(t) = -5q_3(t) - \frac{1}{6}x(t); \quad y(t) = q_1(t) + q_2(t) + q_3(t)$$

$$\begin{bmatrix} q'_1(t) \\ q'_2(t) \\ q'_3(t) \end{bmatrix} = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -5 \end{bmatrix} \begin{bmatrix} q_1(t) \\ q_2(t) \\ q_3(t) \end{bmatrix} + \begin{bmatrix} 1 \\ -1 \\ -\frac{1}{6} \end{bmatrix} x(t)$$

$$y(t) = \begin{bmatrix} 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} q_1(t) \\ q_2(t) \\ q_3(t) \end{bmatrix}$$

Q.3. (a) Obtain expression for current $i(t)$ and voltage across the inductor in the circuit shown in figure. (4)

Ans



$$R_2 i_1(t) - i(t) + R_1 i_1(t) = 20 u(t), \quad 16 [i_1(t) - i(t)] + 4 i_1(t) = 20 u(t)$$

Take LT on both sides.

$$16 [I_1(s) - I(s)] + 4 I_1(s) = \frac{20}{s}$$

$$20 I_1(s) - 16 I(s) = \frac{20}{s} \quad \dots(1)$$

$$R_2 i(t) + L \frac{di(t)}{dt} + R_2 [i(t) - i_1(t)] = 0$$

$$6 i(t) + 0.1 \frac{di(t)}{dt} + 16 [i(t) - i_1(t)] = 0$$

$$6 I(s) + 0.1 s I(s) + 16 [I(s) - I_1(s)] = 0$$

$$0.1 s I(s) + 22 I(s) - 16 I_1(s) = 0 \quad \dots(2)$$

$$I_1(s) = \left[\frac{20}{s} + 16 I(s) \right] \times \frac{1}{20}, \quad I_1(s) = \frac{1}{s} + \frac{4}{5} I(s)$$

$$0.1 s I(s) + 22 I(s) - 16 \left[\frac{1}{s} + \frac{4}{5} I(s) \right] = 0$$

$$0.1 s I(s) + 22 I(s) - \frac{16}{s} - \frac{64}{5} I(s) = 0$$

$$I(s) \left[0.1 s + 22 - \frac{64}{5} \right] - \frac{16}{s} = 0$$

$$I(s) [0.1 s + 9] - \frac{16}{s} = 0$$

$$0.1 s I(s) + 9 I(s) = \frac{16}{s}$$

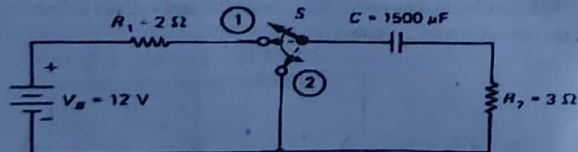
$$0.1 s^2 I(s) + 9 s I(s) - 16 = 0$$

$$I(s) = \frac{1}{(s+45)}$$

Take Inverse LT

$$i(t) = e^{-45t} \quad \text{Ans.}$$

Q.3. (b) In the circuit shown in figure, switch s is off until $t = 0$, moved to position 1 at $t = 0$, and then moved to position 2 at $t = 10\text{ms}$. Determine and sketch the voltage across capacitor.



Ans. Time constant of the circuit is

$$\tau_c = CR = 1500 \times 10^{-6} \times 5 = 7500 \times 10^{-6} = 7.5 \text{ msec}$$

$$\text{for } t \leq 10 \text{ ms} = i_0 e^{-t/\tau_c}$$

$$i_0 = \frac{12}{3} = 4 \text{ A}$$

$$i = 4 e^{-10 \times 10^{-3} / 7.5 \times 10^{-3}}$$

$$i = 4 e^{-1.33 t} \text{ Amp}$$

At this moment potential difference across capacitor is

$$v_c = (1 - e^{-1.33 t}) \text{ Volt}$$

Q.4. (a) Explain initial and final value theorem of Laplace Transform. (2)

Ans. Initial value theorem

$$\lim_{s \rightarrow \infty} L \left[\frac{df(t)}{dt} \right] = f(0^+) = \lim_{s \rightarrow \infty} [s f(s)]$$

Final value Theorem

$$\lim_{s \rightarrow 0} L \left[\frac{df(t)}{dt} \right] = f(\infty) = \lim_{s \rightarrow 0} [s f(s)]$$

Q.4. (b) Determine the current $i(t)$ in a series RLC circuit consisting of $R = 6 \Omega$, $L = 1 \text{ mH}$, $C = 1/8 \mu\text{F}$, when each of source is applied

(a) ramp voltage $6r(t-1)$ and (b) step voltage $2u(t-2)$.

Assume circuit is initially relaxed. (8)

$$\text{Ans. } R i(t) + \frac{1}{C} \int_0^t i(t) dt + L \frac{di(t)}{dt} = v_s(t)$$

Take Laplace Transform on both sides.

$$6 i(t) + 8 \int_0^t i(t) dt + 1 \frac{di(t)}{dt} = 6 r(t-1)$$

$$6 I(s) + 8 \times 10^{-6} \frac{I(s)}{s} + 10^{-3} s I(s) = \frac{6e^{-s}}{s^2}$$

$$\frac{6sI(s) + 8 \times 10^{-6} I(s) + 10^{-3} s^2 I(s)}{s^2} = \frac{6e^{-s}}{s^2}$$

$$6s^2 I(s) + 8 \times 10^{-6} s I(s) + 10^{-3} s^3 I(s) - 6e^{-s} = 0$$

$$I(s) [6s^2 + 8 \times 10^{-6} s + 10^{-3} s^3] = 6e^{-s}$$

$$I(s) = \frac{6e^{-s}}{10^{-3} s^3 + 6s^2 + 8 \times 10^{-6} s} = \frac{6e^{-s}}{s [10^{-3} s^2 + 6s + 8 \times 10^{-6}]}$$

$$= \frac{6000 e^{-s}}{s [s^2 + 6000s + 0.008]}$$

$$I(s) = \frac{6000 e^{-s}}{(s+3000)(s+0.0008)}$$

$$R i(t) + \frac{1}{C} \int_0^t i(t) dt + L \frac{di(t)}{dt} = 2 u(t-2)$$

$$6 \dot{i}(t) + 8 \int_0^t i(t) dt + 10^{-3} \frac{di(t)}{dt} = 2 u(t - 2)$$

$$6 I(s) + 8 \times 10^{-6} \frac{I(s)}{s} + 10^{-3} s I(s) = \frac{2e^{-2s}}{s}$$

$$\frac{6s I(s) + 8 \times 10^{-6} I(s) + 10^{-3} s^2 I(s)}{s} = \frac{2e^{-2s}}{s}$$

$$10^{-3} s^2 I(s) + 6s I(s) + 8 \times 10^{-6} I(s) = 2e^{-2s}$$

$$I(s) [10^{-3} s^2 + 6s + 8 \times 10^{-6}] = 2e^{-2s}$$

$$I(s) = \frac{2e^{-2s}}{10^{-3} s^2 + 6s + 8 \times 10^{-6}}$$

$$I(s) = \frac{2000e^{-2s}}{s^2 + 6000s + 0.008}$$

$$I(s) = \frac{2000e^{-2s}}{(s + 3000)(s + 3000)}$$

END TERM EXAMINATION [JUNE-2024]

FOURTH SEMESTER [B.TECH]

CIRCUITS AND SYSTEMS [EEC-208]

Time: 3 Hrs.

Max. Marks: 75

Note: Attempt five questions in all including Q. No. 1 which is compulsory. Select one question from each unit.

Q.1. Attempt all questions:

Q.1. (a) Find the Fourier transform for $x(t) = \sin(\omega_c t) u(t)$ (3)

Ans.

$$x(t) = \sin(\omega_c t) u(t)$$

$$\sin \omega_c t = \frac{1}{2j} [e^{+j\omega_c t} - e^{-j\omega_c t}]$$

$$\text{FT} [\sin \omega_c t] = \frac{1}{2j} \{ \text{FT} [e^{+j\omega_c t}] - \text{FT} [e^{-j\omega_c t}] \}$$

$$X(\omega) = \frac{1}{2j} [2\pi \delta(\omega - \omega_c) - 2\pi \delta(\omega + \omega_c)]$$

$$X(\omega) = \frac{j\pi}{2} [\delta(\omega - \omega_c) - \delta(\omega + \omega_c)]$$

$$\sin \omega_c t = j\pi [\delta(\omega + \omega_c) - \delta(\omega - \omega_c)]$$

Q.1. (b) List the properties of ROC. (3)

Ans. • The ROC is a ring, whose centre is at origin.

• ROC cannot contain any pole.

• The ROC must be a connected region.

• If ROC of $X(z)$ includes unit circle then only the Fourier transform of DT sequence $x(n)$ converges.• For a finite duration sequence $x(n)$: ROC is entire z -plane except $z = 0$ and $z = \infty$.• If $x(n)$ is causal then ROC is the exterior part of the circle having radius 'a' and for anti-causal ROC is the interior part of the same.

Q.1. (c) Determine the impulse response of series and parallel RL circuit. (3)

Ans. Refer to Q. 2 (b) End Term Examination Dec. 2017. (Pg. No. 10-2017).

Q.1. (d) Explain the AC circuit containing the inductor only. (3)

Ans. Apply KVL

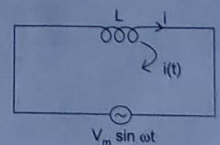
$$V_m \sin \omega t = L \frac{di}{dt}$$

$$di = \frac{V_m}{L} \sin \omega t dt$$

$$i = \frac{V_m}{L} \int \sin \omega t dt$$

$$i = \frac{V_m}{\omega L} (-\cos \omega t) + c$$

$$i = \frac{V_m}{\omega L} \sin \left(\omega t - \frac{\pi}{2} \right)$$



Q.1. (c) Explain short circuit and open circuit impedance in terms of ABCD parameters. (3)

Ans.

$$z_{11} = \frac{A}{C} \quad z_{12} = \frac{AD - BC}{C}$$

$$z_{21} = \frac{1}{C} \quad z_{22} = \frac{D}{C}$$

$$y_{11} = \frac{D}{B} \quad y_{12} = \frac{-\Delta T}{B}$$

$$y_{21} = \frac{-1}{B} \quad y_{22} = \frac{A}{B}$$

UNIT-I

Q.2. Examine the following system: Linear/Nonlinear, Causal/Non-Causal, Time Variant/Time-Invariant, and Static/Dynamic with examples. (15)

Ans. Refer to Q. 2 (a) End Term Exam July 2023. (Pg. No. 9-2023).

Q.3. (i) Find the laplace transform for following signals if $x(t) = e^{-4t} u(t)$. (15)

(i) $y(t) = x(t+2)$; (ii) $y(t) = t \cdot x(t)$; (iii) $y(t) = dx(t)/dt$.

Ans. (i)

$$x(t) = e^{-4t} u(t)$$

$$y(t) = x(t+2)$$

$$y(t) = e^{-4(t+2)} u(t+2)$$

$$F(s) = \int_0^{\infty} e^{-4(t+2)} u(t+2) e^{-st} dt = \int_0^{\infty} e^{-4t} e^{-8} u(t+2) e^{-st} dt$$

$$= e^{-8} \int_{-2}^{\infty} e^{-4t} e^{-st} dt = e^{-8} \int_{-2}^{\infty} e^{-(s+4)t} dt = e^{-8} \left[\frac{e^{-(s+4)t}}{-(s+4)} \right]_{-2}^{\infty}$$

$$= \frac{-e^{-8}}{(s+4)} \left[\frac{1}{\infty} - \frac{1}{e^{-(s+4) \cdot 2}} \right] = \frac{-e^{-8}}{(s+4)} \left[0 - \frac{1}{e^{-2(s+4)}} \right] = \frac{-e^{-8}}{(s+4)} \cdot \frac{e^{-2(s+4)}}{e^{-2(s+4)}} = \frac{-e^{-8-2(s+4)}}{(s+4)}$$

$$F(s) = \frac{e^{-2s}}{s+4}$$

(ii)

$$y(t) = t x(t)$$

$$y(t) = t e^{-4t} u(t)$$

$$f(s) = \int_0^{\infty} t e^{-4t} e^{-st} dt = \int_0^{\infty} t e^{-(s+4)t} dt = t \left[\frac{e^{-(s+4)t}}{-(s+4)} \right]_0^{\infty} - \int_0^{\infty} \frac{e^{-(s+4)t}}{-(s+4)} dt$$

$$= t \left[\frac{e^{-(s+4)t}}{-(s+4)} \right]_0^{\infty} - \frac{1}{(s+4)} \left[-\frac{e^{-(s+4)t}}{s+4} \right]_0^{\infty}$$

$$= \frac{1}{(s+4)} \left[(-1+0) - \left(\frac{1}{s+4} - \frac{1}{s+4} \right) \right]$$

$$F(s) = \frac{-1}{s+4}$$

(iii)

$$y(t) = \frac{dx(t)}{dt}$$

$$x(t) = e^{-4t} u(t)$$

$$y(t) = \frac{d}{dt} [e^{-4t} u(t)]$$

$$y(t) = -4 e^{-4t} \delta(t)$$

$$F(s) = \int_0^{\infty} -4 e^{-4t} e^{-st} dt = -4 e^{-4(s+1)t} \Big|_0^{\infty}$$

$$F(s) = -4$$

UNIT-II

Q.4. (a) Derive the Transient Response in series R-L-C circuit with sinusoidal excitation for second order circuit. (7)

Ans. Refer to Q.3 (a) End Term Examination July 2023. (Pg. No. 12-2023).

Q.4. (b) A step voltage of 20 V is applied at $t=0$ in an R-L-C series circuit where $R = 5\Omega$, $L = 1H$ and $C = 1F$. Calculate $i(t)$ with and without Laplace Transform. (8)

Ans.

$$V = 20 \text{ volt}, R = 5\Omega, L = 1H, C = 1F$$

$$I(s) = \frac{1/L}{(s+\alpha)(s+\beta)}$$

where

$$\alpha, \beta = \frac{-R}{2L} \pm \sqrt{\left(\frac{R}{2L}\right)^2 - \frac{1}{LC}} = \frac{-5}{2} \pm \sqrt{\left(\frac{5}{2}\right)^2 - \frac{1}{1}} \Rightarrow \frac{-5}{2} \pm \sqrt{\frac{21}{4}}$$

$$\Rightarrow -2.5 \pm 2.3$$

$$\Rightarrow -0.2, -4.8$$

we assume

$$\alpha = -0.2, \beta = -4.8$$

so,

$$\alpha > \beta$$

Roots are real but not equal.

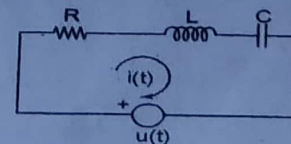
$$i(t) = \left[\frac{1}{L(\beta-\alpha)} e^{-\alpha t} + \frac{1}{L(\alpha-\beta)} e^{-\beta t} \right] \text{ Ampere}$$

$$= -0.28 e^{0.2t} + 0.28 e^{4.8t}$$

$$i(t) = -0.28 (e^{0.2t} - e^{4.8t}) \text{ Ampere}$$

Q.5. (a) Find the step response (by applying a constant voltage source at the close of a switch) of a Parallel RLC circuit. (8)

Ans. Step Response of RLC- Series circuit



$$u(t) = Ri(t) + \frac{1}{C} \int i(t) dt + L \frac{di(t)}{dt}$$

$$\frac{1}{S} = RI(S) + \frac{I(S)}{SC} + SLI(S) = I(S) \left[R + \frac{1}{SC} + SL \right]$$

$$I(S) = \frac{1}{S \left(R + \frac{1}{SC} + SL \right)} = \frac{1}{SR \frac{1}{C} + S^2 L}$$

$$= \frac{1}{L \left(S^2 + \frac{SR}{L} + \frac{1}{LC} \right)} = \frac{1/L}{S^2 + S \frac{R}{L} + \frac{1}{LC}}$$

$$\alpha, \beta = \frac{-R/L \pm \sqrt{\left(\frac{R}{L}\right)^2 - \frac{4}{LC}}}{2}$$

$$\alpha, \beta = \frac{-R}{2L} \pm \sqrt{\left(\frac{R}{2L}\right)^2 - \frac{1}{LC}}$$

Case I: Both α and β are real and not equal i.e. $\frac{R}{2L} > \frac{1}{\sqrt{LC}}$ (overdamped)

$$I(S) = \frac{1/L}{(S+\alpha)(S+\beta)} = \frac{K_1}{S+\alpha} + \frac{K_2}{S+\beta}$$

where

$$K_1 = \frac{1/L}{S+\beta} \Big|_{S=-\alpha} = \frac{1}{L(\beta-\alpha)}$$

$$K_2 = \frac{1/L}{S+\alpha} \Big|_{S=-\beta} = \frac{1}{L(\alpha-\beta)}$$

$$= \frac{1}{L(\beta-\alpha)} + \frac{1}{L(\alpha-\beta)}$$

$$= \frac{1}{S+\alpha} + \frac{1}{S+\beta}$$

$$i(t) = \left[\frac{1}{L(\beta-\alpha)} e^{-\alpha t} + \frac{1}{L(\alpha-\beta)} e^{-\beta t} \right] \text{ Ampere.}$$

Case II: α & β are equal to each other [$\alpha = \beta = \gamma$].

i.e., $\frac{R}{2L} = \frac{1}{\sqrt{LC}}$ Critically Daped

$$I(S) = \frac{1/L}{(S+\gamma)^2}$$

$$I(t) = \frac{1}{L} t e^{-\gamma t} \text{ Ampere.}$$

Case III. $\left(\frac{R}{2L}\right) < \frac{1}{\sqrt{LC}}$ and $\alpha = \beta^*$ Under Damped (oscillatory)

Let $\alpha = A_0 - jB$ and $\beta^* = A_0 + jB$

$$\text{Let } |A_0| = \frac{R}{2L} \quad |B| = \sqrt{\frac{1}{LC} - \left(\frac{R}{2L}\right)^2}$$

$$I(S) = \frac{1/L}{(S+A_0-jB)(S+A_0+jB)} = \frac{K_3}{(S+A_0-jB)} + \frac{K_4}{(S+A_0+jB)}$$

$$K_3 = \frac{1/L}{S+A_0+jB} \Big|_{S=-A_0-jB} = \frac{1}{(2jB)L}$$

$$K_4 = \frac{-1}{(2jB)L}$$

$$I(S) = \frac{1}{(2jB)L} \frac{1}{(S+A_0-jB)} + \frac{-1}{(2jB)L} \frac{1}{(S+A_0+jB)}$$

$$i(t) = \frac{1}{(2jB)L} e^{-A_0 t} e^{jBt} - \frac{1}{(2jB)L} e^{-A_0 t} e^{-jBt} = \frac{e^{-A_0 t}}{BL} \left[\frac{e^{jBt} - e^{-jBt}}{2j} \right]$$

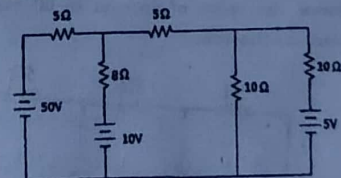
$$i(t) = \frac{e^{-A_0 t}}{BL} \sin Bt \quad \text{Ampere.}$$

Q.5. (b) Determine the step response of series and parallel RC circuits. (7)

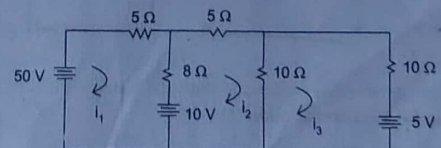
Ans. Refer to Q.3. (b) Mid Term Examination 2023 (Pg. No. 4-2023).

UNIT-III

Q.6. For the circuit shown below, determine the current in each branch using Nodal and Loop analysis. (15)



Ans.



$$50 = 5i_1 + 8(i_1 - i_2) + 10$$

$$40 = 13i_1 - 8i_2 \quad \dots(1)$$

$$\begin{aligned}
 5i_2 + 10(i_2 - i_3) - 10 + 8(i_2 - i_1) &= 0 \\
 5i_2 + 10i_2 - 10i_3 - 10 + 8i_2 - 8i_1 &= 0 \\
 -8i_1 + 23i_2 - 10i_3 &= 10 \quad \dots(2) \\
 10i_3 + 5 \cdot 10(i_3 - i_2) &= 0 \\
 -10i_2 + 20i_3 &= -5 \quad \dots(3)
 \end{aligned}$$

$$\mathbf{I} = \begin{bmatrix} 13 & -8 & 0 \\ -8 & 23 & -10 \\ 0 & -10 & 20 \end{bmatrix} \quad \mathbf{V} = \begin{bmatrix} 40 \\ 10 \\ -5 \end{bmatrix}$$

$$i_1 = \frac{\begin{vmatrix} 40 & -8 & 0 \\ 10 & 23 & -10 \\ -5 & -10 & 20 \end{vmatrix}}{\text{Determinate of I}} = \frac{15600}{3400} = 4.6 \text{ A}$$

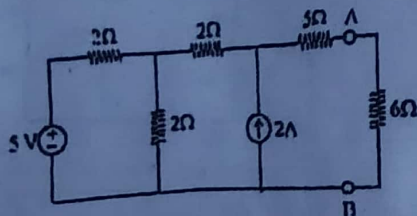
Determinate of I

$$i_2 = \frac{\begin{vmatrix} 13 & 40 & 10 \\ -8 & 10 & -10 \\ 0 & -5 & 20 \end{vmatrix}}{\text{Determinate of I}} = \frac{8350}{3400} = 2.5 \text{ A}$$

$$i_3 = \frac{\begin{vmatrix} 13 & -8 & 40 \\ -8 & 23 & 10 \\ 0 & -10 & -5 \end{vmatrix}}{\text{Determinate of I}} = \frac{3325}{3400} = 0.9 \text{ A}$$

Current in 5 Ω branch is = 4.6 A; Current in 8 Ω branch is = 2.1 A
Current in 10 Ω branch is = 1.6 A

Q.7. Determine the value of current in 6Ω resistor using Thevenin's theorem and Norton's theorem.



Ans. Let us remove 6Ω resistor and short the A B terminal
Apply nodal analysis at node (1)

$$\frac{V-5}{2} + \frac{V}{2} + \frac{V}{2} - 2 + \frac{V}{2} = 0$$

$$\frac{V}{2} - 2.5 + \frac{V}{2} + \frac{V}{2} - 2 + \frac{V}{2} = 0$$

$$\frac{3V}{2} + \frac{V}{5} = 4.5$$

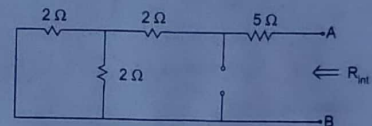
$$\frac{15V + 2V}{10} = 4.5 \Rightarrow 17V = 45$$

$$V = \frac{45}{17} = 2.65 \text{ volt}$$

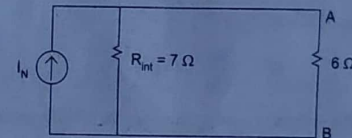
Apply KVL at extreme right loop.

$$-V + I_{s.c.} \times 5 = 0$$

$$I_N = I_{s.c.} = \frac{2.65}{5} = 0.53 \text{ A}$$



$$R_{int} = [(2 \parallel 2) + 1] + 5 = 1 + 1 + 5 = 7 \Omega$$



$$I_{6\Omega} = I_N \times \frac{R_{int}}{R_{int} + 6} = 0.53 \times \frac{7}{7+6} = 0.29 \text{ Ampere}$$

UNIT-IV

Q.8. (a) Explain the term 'tie-set' and 'tie-set matrix' of a network with an illustrative example. (5)

Ans. Refer to Q. 6 (a) End Term Examination Dec. 2017. (Pg. No. 16-2017).

Q.8. (b) Derive and prove the following Matrix Equation with an appropriate example $\mathbf{I}_b = \mathbf{B}^T \mathbf{I}_L$

Where, B is the Tie-set matrix, $\mathbf{I}_b$ and $\mathbf{I}_L$ are the branch current respectively (10)

Ans. Since $[\mathbf{A}] [\mathbf{I}_b] = 0$

$$\therefore [\mathbf{A}_T : \mathbf{A}_L] \begin{bmatrix} \mathbf{I}_T \\ \mathbf{I}_L \end{bmatrix} = 0$$

$$\therefore \mathbf{A}_T \mathbf{I}_T + \mathbf{A}_L \mathbf{I}_L = 0$$

$$\text{i.e.} \quad \mathbf{A}_T \mathbf{I}_T = -\mathbf{A}_L \mathbf{I}_L$$

Pre-multiplying with A^{-1}

$$I_T = A^{-1} A_L I_L$$

$$= -(A^{-1} A_L) I_L$$

Now,

$$[I_b] = \begin{bmatrix} I_T \\ I_L \end{bmatrix} = \begin{bmatrix} -(A^{-1} A_L) I_L \\ I_L \end{bmatrix} = \begin{bmatrix} -\frac{(A^{-1} A_L)}{u} \\ 1 \end{bmatrix} I_L$$

$$[I_b] = [B'] [I_L] \quad \text{Proved}$$

Q.9. (a) Derive the transmission parameters of Two-port networks. Also, determine the condition of Reciprocity and symmetry for ABCD parameter. (8)

Ans.

$$\begin{bmatrix} V_1 \\ I_1 \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} V_2 \\ -I_2 \end{bmatrix}$$

$$V_1 = AV_2 + B(-I_2) \quad \dots(1)$$

$$I_1 = CV_2 + D(-I_2) \quad \dots(2)$$

Open circuit at output end, $I_2 = 0$

$$A = \frac{V_1}{V_2}, \quad C = \frac{I_1}{V_2}$$

Short circuit at output end, $V_2 = 0$

$$B = \frac{V_1}{-I_2}, \quad D = \frac{I_1}{-I_2}$$

Condition of Reciprocity and symmetry in ABCD parameters.

Shorting at output end in equation. (1)

$$V_1 = -BI_2$$

$$\frac{I_2}{V_1} = -\frac{1}{B}$$

$$0 = AV_2 - BI_2 \quad \text{and} \quad I_1 = CV_2 - DI_2$$

This gives,

$$I_2 = \frac{AV_2}{B} \quad \text{and} \quad I_1 = CV_2 - \frac{ADV_2}{B}$$

$$= V_2 \left(\frac{BC - AD}{B} \right)$$

$$\frac{I_1}{V_2} = \frac{BC - AD}{B} = \frac{-AD - BC}{B}$$

Condition of Reciprocity is given by

$$AD - BC = 1 \quad \text{or} \quad \begin{vmatrix} A & B \\ C & D \end{vmatrix} = 1$$

Q.9. (b) Explain the parallel and cascade connection of Two-port networks. (7)

Ans. **Series-Parallel Connection:** Two port networks are said to be connected in series parallel if the input ports are connected in series while the output ports are connected in parallel as shown in figure.

Then, the connections require that

$$V_1 = V_{1a} + V_{1b}$$

$$I_1 = I_{1a} = I_{1b}$$

$$V_2 = V_{2a} = V_{2b}$$

$$I_2 = I_{2a} + I_{2b}$$

For network N_a

$$\begin{bmatrix} V_{1a} \\ I_{2a} \end{bmatrix} = \begin{bmatrix} h_{11a} & h_{12a} \\ h_{21a} & h_{22a} \end{bmatrix} \begin{bmatrix} I_{1a} \\ V_{2a} \end{bmatrix}$$

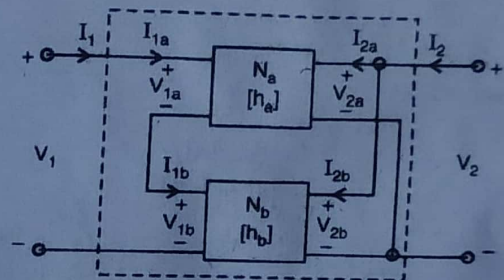


Fig. Series-parallel connection of two-port networks.

Similarly, For network N_b ,

$$\begin{bmatrix} V_{1b} \\ I_{2b} \end{bmatrix} = \begin{bmatrix} h_{11b} & h_{12b} \\ h_{21b} & h_{22b} \end{bmatrix} \begin{bmatrix} I_{1b} \\ V_{2b} \end{bmatrix}$$

Now,

$$\begin{bmatrix} V_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} V_{1a} \\ I_{2a} \end{bmatrix} \begin{bmatrix} V_{1b} \\ I_{2b} \end{bmatrix}$$

$$= \begin{bmatrix} h_{11a} + h_{11b} & h_{12a} + h_{12b} \\ h_{21a} + h_{21b} & h_{22a} + h_{22b} \end{bmatrix} \begin{bmatrix} I_1 \\ V_2 \end{bmatrix}$$

So, in matrix form the h-parameters of the series-parallel connected network can be written as:

$$\begin{bmatrix} V_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} \begin{bmatrix} I_1 \\ V_2 \end{bmatrix}$$

where

$$h_{11} = h_{11a} + h_{11b}$$

$$h_{12} = h_{12a} + h_{12b}$$

$$h_{21} = h_{21a} + h_{21b}$$

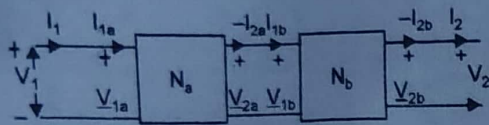
$$h_{22} = h_{22a} + h_{22b}$$

or in the matrix form

$$[h] = [h_a] + [h_b]$$

This result may be generalised for any number of two ports networks connected in series parallel. The overall h-parameter matrix for series-parallel connected two port networks is simply the sum of h-parameter matrices of each individual two-port network connected in series-parallel.

Cascade Connection:



From network 1

$$\begin{bmatrix} V_{1a} \\ I_{1a} \end{bmatrix} = \begin{bmatrix} A_a & B_a \\ C_a & D_a \end{bmatrix} \begin{bmatrix} V_{2a} \\ -I_{2a} \end{bmatrix}$$

From network 2

$$\begin{bmatrix} V_{1b} \\ I_{1b} \end{bmatrix} = \begin{bmatrix} A_b & B_b \\ C_b & D_b \end{bmatrix} \begin{bmatrix} V_{2b} \\ -I_{2b} \end{bmatrix}$$

$$I_1 = I_{1a} - I_{2a} = I_{1b} \quad I_{2b} = I_2$$

$$V_1 = V_{1a} \quad V_{2a} = V_{2b} = V_2$$

$$\begin{bmatrix} V_1 \\ I_1 \end{bmatrix} = \begin{bmatrix} V_{1a} \\ I_{1a} \end{bmatrix} = \begin{bmatrix} A_a & B_a \\ C_a & D_a \end{bmatrix} \begin{bmatrix} V_{2a} \\ -I_{2a} \end{bmatrix} = \begin{bmatrix} A_a & B_a \\ C_a & D_a \end{bmatrix} \begin{bmatrix} V_{1b} \\ I_{1b} \end{bmatrix}$$

$$= \begin{bmatrix} A_a & B_a \\ C_a & D_a \end{bmatrix} \begin{bmatrix} A_b & B_b \\ C_b & D_b \end{bmatrix} \begin{bmatrix} V_{2b} \\ -I_{2b} \end{bmatrix}$$

$$= \begin{bmatrix} A_a & B_a \\ C_a & D_a \end{bmatrix} \begin{bmatrix} A_b & B_b \\ C_b & D_b \end{bmatrix} \begin{bmatrix} V_2 \\ -I_2 \end{bmatrix}$$

...(1)

As we know for any 2 port network 1 parameter equation is

$$\begin{bmatrix} V_1 \\ I_1 \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} V_2 \\ -I_2 \end{bmatrix}$$

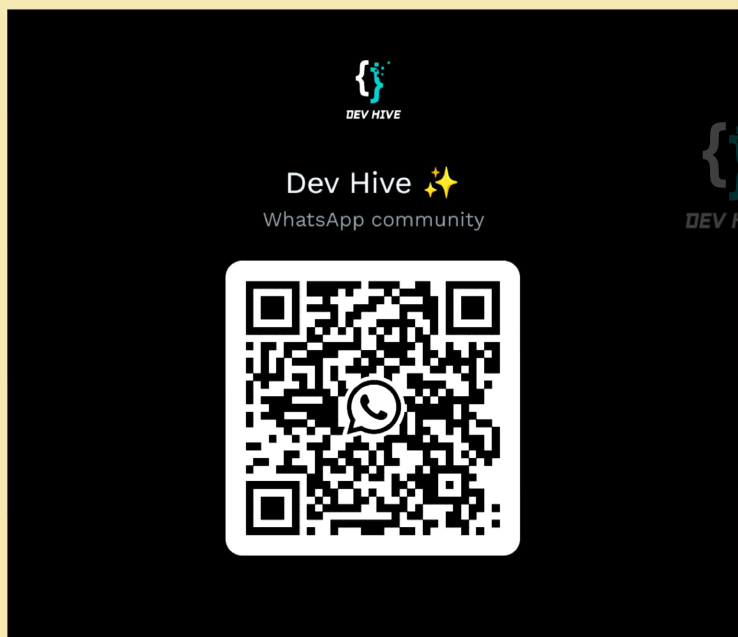
...(2)

On comparing above 2 equation

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} A_a & B_a \\ C_a & D_a \end{bmatrix} \begin{bmatrix} A_b & B_b \\ C_b & D_b \end{bmatrix}$$

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